



Jashore University of Science and Technology
Department of Computer Science and Engineering

Semester Final Examination-2023

2nd Year 1st Semester, Session: 2020-2021

Course Code: CSE 2101

Time: 3.00 hours

Course Title: Object Oriented Programming

Marks: 72

[N.B. Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks.]

1. a) Discuss the basic principles of OOP with examples. 08
b) What are the access modifiers in Java? Write down the purpose of it. 04
2. a) What is class? Discuss the basic structure of a class with an example. 03
b) Describe the benefit of using constructor in a class by using an example. 03
c) What is meant by polymorphism? Explain the different types of polymorphism including the differences between them with example. 06
3. a) "Encapsulation is the most important concept among the OOP concepts" Explain the statement in your own words. 06
b) Describe the association and composition concept with examples in OOP. 06
4. a) Describe the concept of Exception-handling in OOP with examples. 06
b) Differentiate between error and exception. 02
c) Describe the role of exception handling in file I/O with examples. 04
5. a) Create three classes with the following names "Student", "Teacher" and "Officer". Write the attributes and methods of the classes to perform an examination. (Include the attributes and methods based on your real-life examination experience) 06
b) Draw the class-based model from (5a) with proper relationship. Will there be any other classes necessary to complete the overall procedure? Explain your answer. 06
6. a) "Open for Extension and Close for Modification" explain the statement in the light of OOP with examples. 06
b) Explain up-casting and down-casting with appropriate examples. 04
c) Differentiate between abstract class and interface. 02
7. A program which decrypts files under the Swap Encryption Scheme by swapping pairs of characters is given below. Some code has been omitted and you need not to understand the operation of the algorithm.

```
1  class Swapper extends Reader {
2      private final PushbackReader pushBack;
3      Swapper(PushbackReader p) { pushBack = p; }
4
5      @Override
6      public int read(char[] cbuf, int off, int len) {
7          int r = wrap.read(cbuf, off, len);
8          if (r % 2 == 1) { pushBack.unread(cbuf, off + --r, 1); }
9          for (int i = 0; i < r; i += 2) { swap(cbuf, i, i + 1); }
10         return r;
11     }
12 }
13
14 class Decryptor {
15     static List<String> read(String fileName) {
16         try (BufferedReader r = new BufferedReader(new Swapper(
17             new PushbackReader(new FileReader(fileName)))) {
18             return readLines(r);
19         }
20     }
21 }
```

- a) The four principles of object-oriented programming are encapsulation, abstraction, inheritance and polymorphism. Explain how the program makes use of each of them, with reference to specific lines in the code.

8. Analyzing the system for online ticket booking, the following requirements were identified –

1. There can be three types passenger: General, Premium and New
 2. Passenger can register into the system with his – NID, Name, Email, Mobile Number, and Password
 3. Passenger can login into the system with his email and password
 4. Passenger can buy ticket
 5. Ticket should be confirmed by the System Admin
 6. Passenger can pay the ticket price online after being confirmed
 7. Online payment should be using the "PAYBD" payment gateway
 8. Passenger can cancel a booking
 9. Passenger can request for return money for the canceled booking
- a) Identify the classes with their attributes and methods.
- b) Create the Class-based Model with proper connections.

PASS -
system

login

buy

system admin

ticket

Online Payment



Department of Computer Science and Engineering

Jashore University of Science and Technology

2nd Year 1st Semester Final Examination-2023

Session: 2020-21

Course Code: CSE 2103

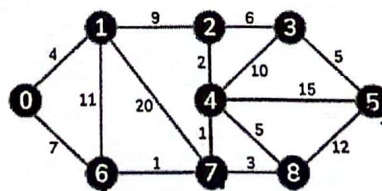
Time: 3.00 hours

Course Title: Algorithm Analysis and Design

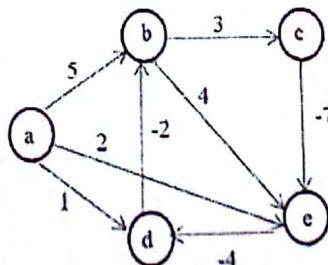
Marks: 72

[N.B. Answer any 6 questions from following 8 questions. The figures in the margin indicate full marks.]

1. a) Why algorithm study is important? What are the steps to develop an algorithm? 2+2 +2
What are the characteristics of a good algorithm.
- b) Elaborate the idea of best case, average case and worst-case scenario of an algorithm with examples. What are the corresponding notions 06
2. a) Sort the following items in descending order using merge sort and quick sort 3+3
500, 600, 100, 23, 152, 712, 900
- b) Identify the space and time complexity for your responses in question 2(a). 06
3. a) Design an algorithm for fractional knapsack problem. Specify at least two shortcomings of your algorithm. What is the space complexity of your problem? 3+2 +1
- b) Discuss the max-min algorithm with its computational advantages over the linear search. 4+2
4. a) Devise the comparative study between dynamic programming and greedy solution 04
- b) Identify the lowest cost and corresponding sequence for the following matrix chain 6+2
10, 15, 2, 5, 15.
Discuss the time complexity of the above algorithm
5. a) Considering the following undirected weighted graph, find the shortest path between the node 0 and 5. Clearly show the interim steps of your process. 08



- b) Design the algorithm/pseudo code of the above response 04
6. a) Discuss the pros and cons of 'Floyd Warshall algorithm' over the other shortest path finding algorithms. Explain a case where 'Floyd Warshall algorithm' will fail to find solution. 3+3
- b) Identify the shortest path between a and c using Floyd Warshall algorithm. Clearly demonstrates the steps 06



7. Consider an algorithm as follows:

```
Bellman-Ford(G,w,s)
// Graph G=(V,E), w:V->R a weight function, s is the
// source vertex
// uses distance estimates d[v] for v in V
// and parents p[v] for v in V with p[s]=s for the source
1 Initialize(G,s)
2 for i = 1 to |V|-1
3   for each edge (u,v) in E do
4     Relax(u,v,w)
5   end
6 end
7 for each edge (u,v) in E do
8   if d[v] > d[u]+w(u,v) then return false
9 end
10 return true
```

- a) Give the code of the procedure Relax(u,v,s) . 05
- b) Derive the complexity of Bellman-Ford, assuming that Initialize(G,s) takes $\Theta(V)$ time. 04

Note: Do not just state the result, but explain what each loop contributes. Use line numbers as references.

- c) What does the loop in lines 7–9 do? Explain in brief. 03

8 Consider a weighted directed graph with 5 vertices and weights on the edges:

$w(1,2) = 3, \quad w(1,3) = 1, \quad w(2,3) = 3, \quad w(2,4) = 2,$
 $w(2,5) = -2, \quad w(3,5) = 5, \quad w(4,2) = 1, \quad w(4,5) = -3$

- a) Compute the minimum cost path from 1 to 5. 04
- b) Explain why Dijkstra's algorithm does not work for this type of problem. 03
- c) What kind of algorithm will you use? Justify your opinion. 05

Illustrate the execution of the algorithm on this graph.

-----The End-----



Jashore University of Science and Technology
Department of Computer Science and Engineering

Semester Final Examination-2023

2nd Year 1st Semester, Session: 2020-2021

Course Code: CSE 2107

Time: 3.00 hours

Course Title: Numerical Analysis

Marks: 72

[N.B. Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks.]

1. a) Write short notes on- 4*2
1. Accuracy
 2. Precision
 3. Round-off errors
 4. Floating-point representation
- b) Determine the range of integers in base 10 that can be represented on a 32-bit computer. 04
2. a) Explain the concept of 2*2
1. Convergence
 2. Divergence.
- b) Determine the root of $f(x) = 0.95x^3 - 5.9x^2 + 10.9x - 6$ 4+4
1. Using Newton-Raphson method (three iterations, $x_i = 3.5$)
 2. Using the secant method (three iterations, $x_{i-1} = 2.5$ and $x_i = 3.5$)
3. a) Determine the real roots of $f(x) = -0.5x^2 + 0.25x + 4.5$ using three iterations of bisection method, initial guesses of $x_l = 5$ and $x_u = 10$. Calculate the estimated error and true error after each iteration. 06
- b) Describe the False-position method with proper figures. 04
- c) Differentiate the Bisection and False-position methods. 02
4. Suppose, you are new in Jashore. You have gathered the following information relating the housing rent of three-bedroom homes with the sizes located nearly at Jashore Boat Club.
- | | | | | |
|------------------------|-------|-------|-------|-------|
| Size (in Square Feet) | 1800 | 2100 | 2400 | 2200 |
| Housing Rent (in Taka) | 13200 | 15400 | 17500 | 14800 |
- a) Considering the above data, explicitly write Lagrange interpolation formula. Also estimate the rent for a house of 2000 square feet (using Lagrange formula). 06
- b) Considering the given data, write a C program by implementing the Lagrange interpolation that takes the size of a house as input and prints the estimated housing rent as output. 06
- a) Use Muller's method to determine the real roots of $f(x) = x^3 + x^2 - 3x - 5$. 06

- b) Divide the polynomial $f(x) = x^5 - 5x^4 + x^3 - 6x^2 - 7x + 10$ by the monomial factor $x - 2$. 06
6. a) Evaluate the polynomial $y = x^3 - 7x^2 + 8x - 0.35$ at $x = 1.37$. Use 3-digit arithmetic with chopping. Evaluate the percent relative error. 06
- b) Explain the Taylor-series with proper examples. How we can use Taylor-series to estimate truncation error. 06
7. a) Consider the following system of equations. 06

$$\begin{array}{rrcr} 1x_1 & + & 3x_2 & + & 2x_3 & = & 2 \\ 2x_1 & + & 7x_2 & + & 7x_3 & = & -1 \\ 2x_1 & + & 5x_2 & + & 2x_3 & = & 7 \end{array}$$

Apply Gauss elimination method to solve the given system. Also comment on the nature of the solution.

- b) Now consider the new system of linear equations. Solve the following system and comment on the nature of the solution by applying Gauss elimination method. 06
8. a) Solve the following system of equations by LU decomposition. 0

$$8x_1 + 4x_2 - x_3 = 11$$

$$-2x_1 + 5x_2 + x_3 = 4$$

$$2x_1 - x_2 + 6x_3 = 7$$

- b) Determine the matrix inverse from (8a).



Department of Computer Science and Engineering
Jashore University of Science & Technology
2nd Year 1st Semester (2020-21) - 2023

Course Code: CSE 2105

Course Title: Computer Architecture

Total Time: 3.00 hours

Total Marks: 72.0

(Note: Answer any six set of questions from the following eight set of questions)

1. (a) What is the concept of Von Neumann machine? 04
(b) How CPU executes a Program? Give detailed description. 08
2. (a) Describe Computer Memory Hierarchy. Why Memory Hierarchy is needed? What is the principle behind it? 06
(b) Differentiate between Unified vs. Split caches. Which one is better? Why? 03
(c) Explain Write Policy of Cache Memory. 03
3. (a) What are the types of error in memory? Describe the correction process with block diagram. 04
(b) Describe Disk Performance Parameters. What happens when a process issues an I/O request? Explain with figure. 04
(c) Explain how a single-error-correcting, double-error-detecting code works. 04
4. (a) Suppose, we read a word from main memory (where data bits are saved along with parity bits generated using the hamming code) and we get 101101001111. Is there any error here? If yes, show correct data bits. 09
(b) Describe different memory accessing methods. 03
5. (a) Describe a typical organization of a 32M 8-bit DRAM. 06
(b) Draw a memory module consisting of 8M 8-bit words. 06
6. (a) Describe the process of write and read in a magnetic disc. 04
(b) Describe the data organization of a magnetic disc. 04
(c) What is RAID? Describe them. 04
7. (a) Consider a machine with a byte addressable main memory of 2^{16} bytes and block size of 8 bytes. Assume that a direct mapped cache consisting of 32 lines is used with this machine.
 - i. How is a 16-bit memory address divided into tag, line number and byte number?
 - ii. Into what line would bytes with each of the following addresses be stored?
0001 0001 0001 1011
1100 0011 0011 0100
1101 0000 0001 1101
1010 1010 1010 1010
 - iii. How many total bytes of memory can be stored in the cache?
(b) What is the difference between DRAM and SRAM in terms of characteristics such as speed, size, and cost? 04
(c) Explain the term striped data. 02
8. (a) What is CPU register? Discuss the role of control and status register in CPU. 04
(b) What is the difference between an arithmetic shift and a logical shift? 02
(c) Give a description of operation performed by the operation name given below. 04
(i) PUSH (ii) LEA (iii) MUL (iv) SHL/SHR
(d) How dynamic memory is refreshed? 02



Department of Computer Science and Engineering
Jashore University of Science & Technology
2nd Year 1st Semester (2020-21) - 2023

Course Code: MATH 2101

Course Title: Matrices and Vectors

Total Time: 3.00 hours

Total Marks: 72.0

(Note: Answer any six (06) set of questions from the following eight (08) set of questions)

1. (a) Describe scalar and vector fields with examples. Determine the angles α, β and γ which the vector $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ makes with the positive directions of the coordinate axes and show that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$. 05
- (b) Define rectangular unit vectors. If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are non-coplanar vectors determine whether the vectors $\mathbf{r}_1 = 2\mathbf{a} - 3\mathbf{b} + \mathbf{c}$, $\mathbf{r}_2 = 3\mathbf{a} - 5\mathbf{b} + 2\mathbf{c}$, and $\mathbf{r}_3 = 4\mathbf{a} - 5\mathbf{b} + \mathbf{c}$ are linearly independent or dependent. 04
- (c) If $\mathbf{r}_1 = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{r}_2 = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$, $\mathbf{r}_3 = -2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$, and $\mathbf{r}_4 = 3\mathbf{i} + 2\mathbf{j} + 5\mathbf{k}$, find scalars a, b, c such that $\mathbf{r}_4 = a\mathbf{r}_1 + b\mathbf{r}_2 + c\mathbf{r}_3$. 03
2. (a) Find an equation for the plane determined by the points $P_1(2, -1, 1)$, $P_2(3, 2, -1)$, and $P_3(-1, 3, 2)$. 03
- (b) Define reciprocal sets of vectors. If $\mathbf{a}' = \frac{\mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}$, $\mathbf{b}' = \frac{\mathbf{c} \times \mathbf{a}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}$ and $\mathbf{c}' = \frac{\mathbf{a} \times \mathbf{b}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}$; prove that $\mathbf{a} = \frac{\mathbf{b}' \times \mathbf{c}'}{\mathbf{a}' \cdot \mathbf{b}' \times \mathbf{c}'}$, $\mathbf{b} = \frac{\mathbf{c}' \times \mathbf{a}'}{\mathbf{a}' \cdot \mathbf{b}' \times \mathbf{c}'}$, $\mathbf{c} = \frac{\mathbf{a}' \times \mathbf{b}'}{\mathbf{a}' \cdot \mathbf{b}' \times \mathbf{c}'}$. 05
- (c) Determine a unit vector perpendicular to the plane of $\mathbf{A} = 2\mathbf{i} - 6\mathbf{j} - 3\mathbf{k}$ and $\mathbf{B} = 4\mathbf{i} + 3\mathbf{j} - \mathbf{k}$. 04
3. (a) Define continuity and differentiability. A particle moves along the curve $x = 2t^2, y = t^2 - 4t, z = 3t - 5$, where t is the time. Find the components of its velocity and acceleration at time $t = 1$ in the direction $\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$. 05
- (b) If $\mathbf{A} = \sin u\mathbf{i} + \cos u\mathbf{j} + u\mathbf{k}$, $\mathbf{B} = \cos u\mathbf{i} - \sin u\mathbf{j} - 3\mathbf{k}$, and $\mathbf{C} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$, find $\frac{d}{du}(\mathbf{A} \times (\mathbf{B} \times \mathbf{C}))$ at $u = 0$. 03
- (c) If $\mathbf{A} = (2x^2y - x^4)\mathbf{i} + (e^{xy} - y\sin x)\mathbf{j} + (x^2\cos y)\mathbf{k}$, find: $\frac{\partial A}{\partial x}, \frac{\partial A}{\partial y}, \frac{\partial^2 A}{\partial x^2}, \frac{\partial^2 A}{\partial y^2}, \frac{\partial^2 A}{\partial x \partial y}, \frac{\partial^2 A}{\partial y \partial x}$. 04
4. (a) Find equations for the tangent plane and normal line to the surface $z = x^2 + y^2$ at the point $(2, -1, 5)$. 04
- (b) If $\nabla U = 2r^4\mathbf{r}$, find U . Prove that the vector $\mathbf{A} = 3y^4z^2\mathbf{i} + 4x^3z^2\mathbf{j} - 3x^2y^2\mathbf{k}$ is solenoidal. 04
- (c) Find the acute angle between the surfaces $xy^2z = 3x + z^2$ and $3x^2 - y^2 + 2z = 1$ at the point $(1, -2, 1)$. 04
5. (a) State the Green's theorem. 02
- (b) If $\vec{F} = (2x^2 - 3z)\vec{i} - 2xy\vec{j} - 4x\vec{k}$, evaluate $\iiint_V (\nabla \times \vec{F}) dV$, where V is the closed region bounded by $x = 0, y = 0, z = 0$ and $2x + 2y + z = 4$. 06
- (c) Find the angle between the vectors $\vec{i} + \vec{j} + \vec{k}$ and $\lambda^2\vec{i} - 2\lambda\vec{j} + \vec{k}$. For what value of λ will the vectors be perpendicular? 04

- 6 (a) Define Hermitian and Skew Hermitian matrices. Show that every square matrix may be written as the sum of a Hermitian and a skew-Hermitian matrices. 03
- (b) Show that the transpose of the product of two matrices is the product in reverse order of their transposes i.e., $(AB)' = B'A'$. 03

- (c) Show that the both matrices $\begin{pmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{pmatrix}$ and $\begin{pmatrix} 4 & 3 & 3 \\ -1 & 0 & -1 \\ -4 & -4 & -3 \end{pmatrix}$ are involutory. 03

- 7 (a) Define upper triangular and lower triangular matrices, symmetric and skew-symmetric matrices and orthogonal matrix. 05

- (b) Find the inverse of the matrix $A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 0 & -1 \\ 3 & 3 & 2 \end{bmatrix}$ 07

Also find the symmetric and skew symmetric parts of the above matrix A .

- 8 (a) Define minor of a matrix. Find the rank of the matrix using minor process

$$A = \begin{bmatrix} 0 & 6 & 6 & 1 \\ -8 & 7 & 2 & 3 \\ -2 & 3 & 0 & 1 \\ -3 & 2 & 1 & 1 \end{bmatrix}$$

- (b) Find the rank of the matrix $A = \begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 3 & 4 & 1 \\ 2 & 3 & 7 & 5 \\ 2 & 5 & 11 & 6 \end{bmatrix}$ by reducing it to canonical form. 07

Department of Computer Science and Engineering

Jashore University of Science and Technology

2nd Year 1st Semester Final Examination-2023

Session: 2020-21

Course Code: HUM 2101

Time: 3.00 hours

Course Title: Sociology and Disaster Management

Marks: 72

[N.B. Answer any 6 questions from following 8 questions. The figures in the margin indicate full marks.]

- | | | | |
|----|----|--|-------|
| 1. | a) | What do you understand by the term 'Government'? What is the difference between Absolute monarchy and Constitutional monarchy? | 2+2 |
| | b) | In a feudal society, states is based on land ownership. What is the political importance of feudalism? | 04 |
| | c) | What are the organs of the government? Differentiate among them. | 1+3 |
| 2. | a) | Define the following terms: | 1×4=4 |
| | | <div style="display: flex; justify-content: space-between;"> <div> i) Budget
ii) Hearing </div> <div> iii) Lobby
iv) Amendment </div> </div> | |
| | b) | What do you understand by the term 'Modern State'? What are the functions of the modern state? | 2+2 |
| | c) | What are the contemporary aspects of public administration regarding the following topics: | 2+2 |
| | | <div style="display: flex; justify-content: space-between;"> <div> i) Protecting and advancing democracy
ii) Ensuring Environmental Sustainability </div> </div> | |
| 3. | a) | What do you mean by social dynamics? | 03 |
| | b) | Juvenile gangs are the result of social unrest. Give your opinion. | 04 |
| | c) | What are the impacts of technological factors in real life disaster management risk reduction? | 05 |
| 4. | a) | What do you understand by the term 'Sociology'? What are the significant features of sociology? | 2+2 |
| | b) | What is 'Social Structure'? To understand the social structure, we have to understand different components. What are those components of social structure? | 1+3 |
| | c) | Define following terms: | 1×4=4 |
| | | <div style="display: flex; justify-content: space-between;"> <div> i. Globalization
ii. Bureaucracy </div> <div> iii. Caste
iv. Human Ecology </div> </div> | |
| 5. | a) | Differentiate between: | 2+2 |
| | | <div style="display: flex; justify-content: space-between;"> <div> i.) Quantitative data and Qualitative data
ii.) Open-ended questions and close-ended questions. </div> </div> | |
| | b) | What do you understand by the term 'Research'? To do any kind of social research, you have to identify a study area. What kinds of issues do you have to consider before selecting an area as you study area for research. | 1+2 |

P.T.O.

- c) What are the implications of the following research methods: 2+2
- i.) Ethnography
 - ii.) Participant observation
- d) What is the meaning of 'Community' to you? 01
6. a) Culture is the systems of knowledge shared by a relatively large group of people. But sometimes, cultural lag happens. What do you understand by the term 'cultural lag'? 02
- b) What does a society become complex? State this aspect mentioning the components of a complex society. 1+2
- c) What is capitalism? What do you know about the growth of capitalism between the time 1600 and 1700s? 1+3
- d) Differentiate between material and non-material culture. 03
7. a) "Nuclear families have a functional fit with industrial society"- Justify the statement. 03
- b) What are the relationships between industrial society and the development of urbanization? 03
- c) Sketch the following framework: 2+2
- i.) Disaster Management institutional framework
 - ii.) Disaster Management Regulatory framework
- d) What are the difference between the nuclear family and the extended family? 02
8. a) The population is the whole number of inhabitants occupying an area and continually being modified by increases and losses. What are the factors which affect population change? 03
- b) What are pull and push factors of migration. 02
- c) What is the future of technology in disaster management? 03
- d) What types of technologies in disaster risk reduction can be used to ensure more effective and creative disaster management? 04

-----The End-----

Department of Computer Science & Engineering

Jashore University of Science and Technology

2nd year 1st Semester, B.Sc. Engineering Final Examination, 2022 Session (2019-20)

Course Title: Object Oriented Programming

Time: 3 Hours

Course code: CSE 2103

Full marks: 72.0

(Answer any six set of questions from the followings)

1. a. Distinguish between the following terms and provide suitable examples for each:

8

- i) Objects and Classes ii) Data abstraction and data encapsulation
iii) Inheritance and polymorphism iv) Dynamic binding and message passing

b. What will be output of the below code? Write answer with explanation.

4

i) <pre>public class A { public static void main (String[] args) { if (5) break; } }</pre>	ii) <pre>public class A { public static void main (String[] args) { System.out.println('j' + 'a' + 'v' + 'a'); } }</pre>
iii) <pre>public class A { public static void main (String[] args) { int \$_ = 5; } }</pre>	iv) <pre>public class Demo{ public static void main (String[] arr){ } public static void main (String arr){ } }</pre>

a. Differentiate between Object Oriented Programming and Procedural Programming.

b. When do we use a protected access modifier? How can we prevent a class from being inherited by other classes?

c. What is the output of the following program? Provide reason(s) for your answer.

<pre>public class Vehicle { int id = 120; public int getId() { return id; } }</pre>	<pre>public class Car extends Vehicle { int id = 100; public int getId() { return(id - 20); } public static void main(String[] args) { Vehicle vc = new Car(); Car car = (Car) vc; System.out.print(vc.id + ", " + vc.getId() + ", "); System.out.print(car.id + ", " + car.getId()); } }</pre>
--	---

What are the benefits of Encapsulation in OOP? Do you think encapsulation is properly ensured in the

following class? If not, please modify the code and ensure it.

```
public class Person {
    public int age;
    public string name;
    public string address;
}
```

- b. Explain the Is-a Relationships and Has-a Relationships in OOP. Let's say a system contains the following objects. Can you identify the relationship among these objects?

1. Bike	2. Plusar	3. Engine
4. Building	5. Window	6. Door
7. Student	8. Teacher	9. Person

4. a. What is the signature of a method?

b. In the below Class X, is 'method' properly overloaded?	c. What will be the output of the following program?
<pre>class X { int method(int i, int d) { return i+d; } static int method(int i, double d) { return (int)(i+d); } double method(double i, int d) { return i+d; } static double method(double i, double d) { return i+d; } }</pre>	<pre>class X { void method(int a) { System.out.println("ONE"); } void method(double d) { System.out.println("TWO"); } } class Y extends X { @Override void method(double d) { System.out.println("THREE"); } } public class MainClass { public static void main(String[] args) { new Y().method(100); } }</pre>

d. What are the differences between abstract class and interface?

6

5. a. CSE, JUST wants to develop a **Canteen Management System**. Identify and implement possible classes of this system. Use inheritance if there are similar properties among the classes. You can use abstract class, interface and other oop components if necessary.

12

7. a. Write a program using any language that implements an abstract class **Animal** that has a **Name** property of type **text** and three methods **SetName** (string name), **GetName** and **Eat**. The **Eat** method will be an abstract method of type **void**.

7

You will also need to create a **Dog** class that implements the above **Animal** class and the **Eat** method that says the dog is **Eating**.

To test the program ask the user for a dog name and create a new **Dog** type object from the **Main** of the program, give the **Dog** object a name, and then execute the **GetName** and **Eat** methods.

b. What are the differences between association, aggregation and composition?

5

7. a. The **Collections** class of **Java Collection Framework** has several methods (e.g. **sort()**, **min()**, **max()**) that require natural ordering of the classes to be handled. Explain the two possible conditions that must be satisfied on the classes to be able to use the **sort()** method to arrange the list of objects these classes.

5

b. Explain why you would choose a **LinkedHashSet** (or **LinkedHashMap**) over a **HashSet** (or **HashMap**, respectively).

3

c. Distinguish between

4

i) **InputStream** and **Reader** classes

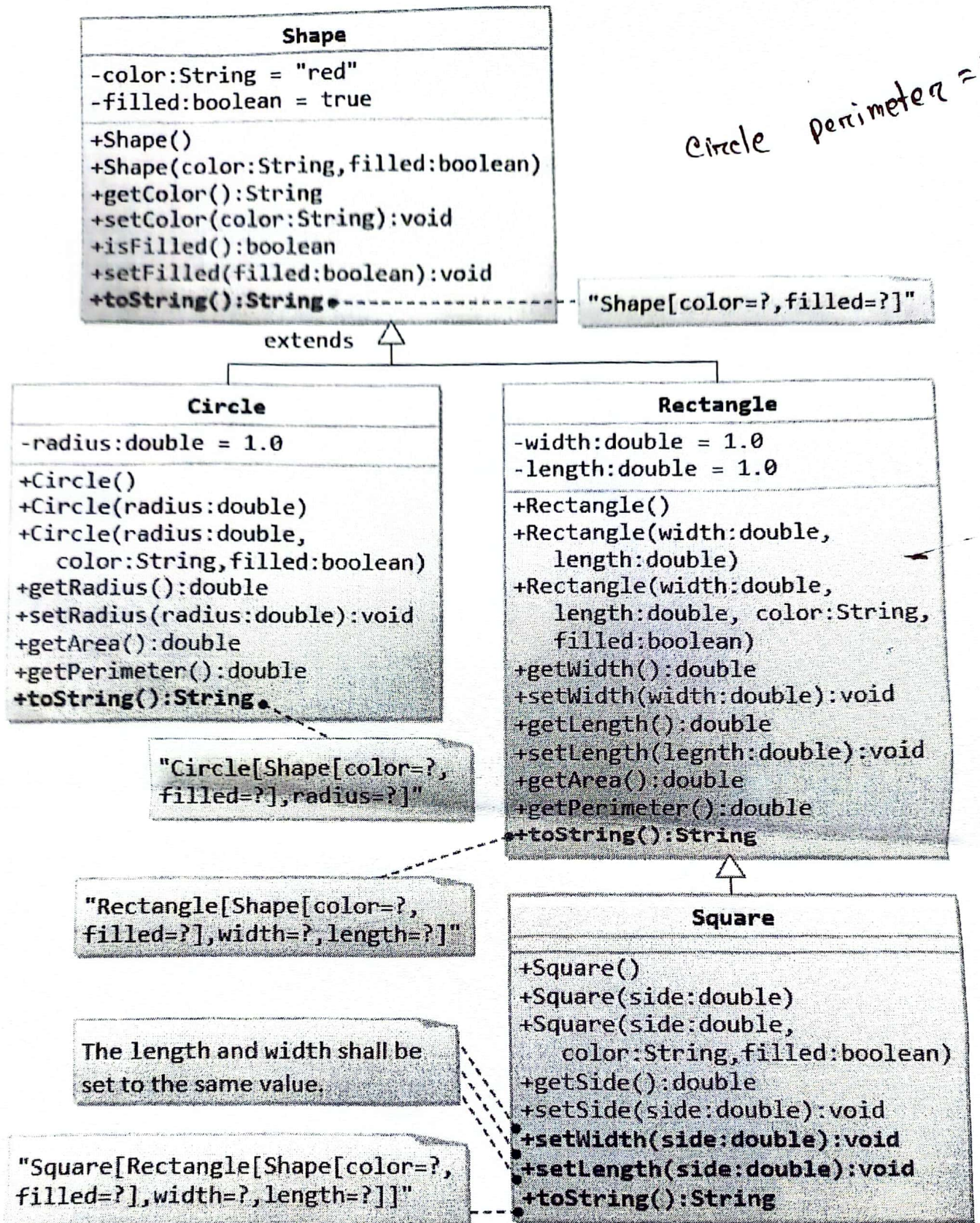
ii) **OutputStream** and **Writer** classes

a. Write a superclass called **Shape** (as shown in the class diagram), which contains:

1

1. Two instance variables **color** (**String**) and **filled** (**boolean**).
2. Two constructors: a no-arg (no-argument) constructor that initializes the **color** to "green" and **filled** to true, and a constructor that initializes the **color** and **fills** to the given values.
3. Getter and setter for all the instance variables. By convention, the getter for a boolean variable **xxx** is called **isXXX()** (instead of **getXxx()** for all the other types).
4. A **toString()** method that returns "A Shape with color of xxx and filled/Not filled".

6m2.





Course Code: CSE 2101

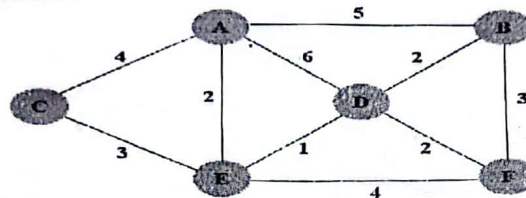
Course Title: Introduction to Algorithms

[N.B. Answer any 6 questions from following 8 questions. The figures in the margin indicate full marks.]

Time: 3.00 hours

Marks: 72

1. a) How can you measure the efficiency of an algorithm? Describe Euclid's algorithm. 05
b) Define $\Omega(n)$ and $\Theta(n)$. Show that following equalities are correct 04
i. $n^{k+e} + n^e \log n = \Theta(n^{k+e})$ ii. $n^3 + 10^6 n^2 = \Theta(n^3)$
c) Define binary search. From the array find the positions of -6, and 125 using binary search. 03
Show the steps. -15, -6, 0, 7, 9, 23, 54, 82, 110, 120
2. a) Write how divide and conquer method works. Write an algorithm to find the maximum and minimum from a list of integers using divide and conquer method. 07
b) Show the steps used to sort the following data using merge sort technique in ascending order: 310, 285, 179, 652, 351, 423, 861, 254, 450 and 520. 05
3. a) Define Greedy Algorithms and Write the elements of the greedy strategy. 03
b) What is an optimal Huffman code for the following set of frequencies? Show fixed and variable length bits/character. 04
a:1, b:1, c:2, d:3, e:5, f:8, g:13, h:21
c) What is MST? Find minimum spanning tree by applying prim's algorithms on following weighted graph where the root vertex is A 05

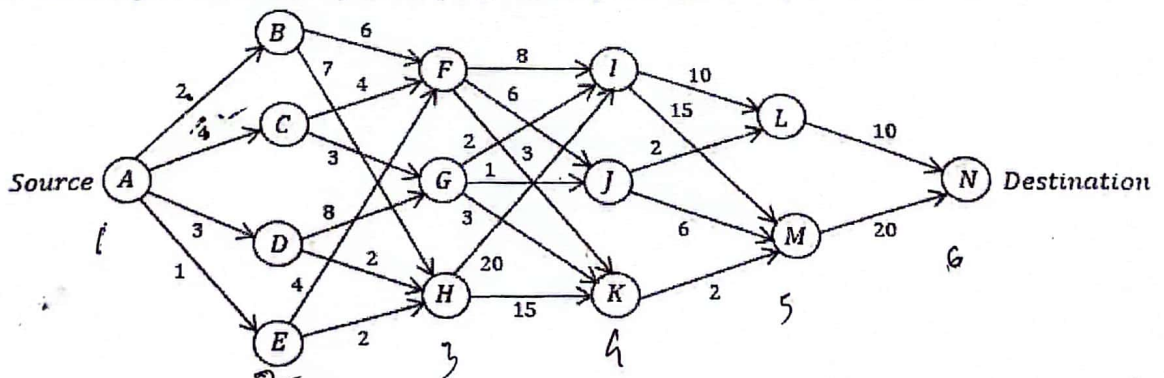


4. a) Consider the following set S of activities: 04

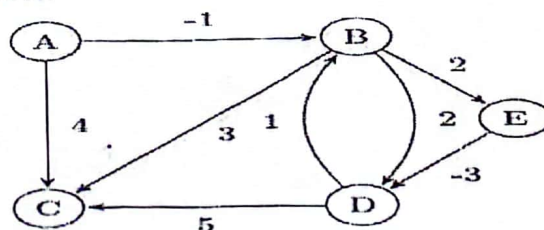
S_i	0	1	2	3	3	5	5	6	8	8	12
f_i	6	4	14	5	9	7	9	10	11	12	16

Now sketch the operation of recursive activity selector algorithm and write the greedy activity selector algorithm.

- b) From the figure show that dynamic programming is better than greedy method. 06

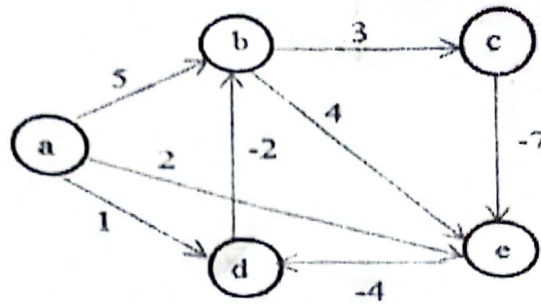


- c) Define Dynamic Programming and write the sequence of steps to develop a dynamic programming algorithm. 02
5. a) Find out the Longest Common subsequences for the following sequences: 04
X: BCDAACD
Y: ACDBAC
b) Show the executions of Bellman-Ford algorithms for the following weighted directed graph where the source vertex is A. 0



- c) What is all pair shortest path problem? Show an example.

6. a) Find the sequence of matrices $D^{(k)}$ computed by the Floyd-Warshall algorithm for the following weighted directed graph.



- b) Write the differences between BFS and DFS. 03
 c) Describe the recursive top down and bottom up rod cutting algorithm. 04
7. a) For a binary tree traversal, if the InOrder and PreOrder traversal show the following node sequences, then find the node sequence for the PostOrder method after constructing the tree. 06
 InOrder: 4, 2, 5, 1, 3, 6
 PreOrder: 1, 2, 4, 5, 3, 6
- b) From the multistage graph of fig. 7(a) show that dynamic programming is better than greedy method. 06

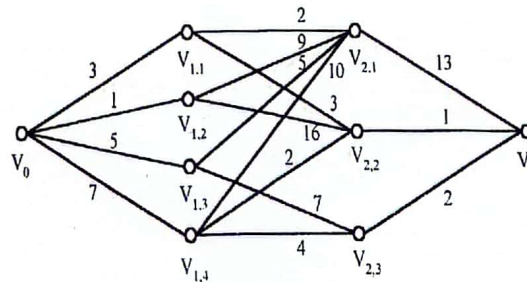
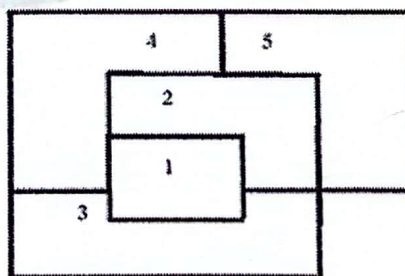


Fig. 7(a)

8. a) What is Backtracking? Describe backtracking solution of the 4-queen problem by drawing the chess board and state space tree.
 b) What is planner graph? Represent the following map into a planner graph. How many colors are needed to color it?



- c) "All NP-complete problems are NP-hard, but some NP-hard problems are not NP-complete"-Justify.

Department of Computer Science and Engineering
Jashore University of Science & Technology
2nd Year 1st Semester Final Exam-2022 (Session: 2019-20)

Course Code: CSE-2105

Course Title: Computer Architecture and Organisation

Total Time: 3.00 hours

Total Marks: 72.0

(Note: Answer any six set of questions from the following eight set of questions)

- | | | |
|---|---|----|
| 1 | (a) What is the concept of Von Neumann machine? | 4 |
| | (b) Why Memory Hierarchy is needed? How the hierarchy is organized? | 8 |
| 2 | (a) What are the Method of Accessing of Computer Memory? Explain the process of Random Access. | 6 |
| | (b) Why Cache Memory is needed? What is the 'Principle' behind a Cache Memory? | 6 |
| 3 | (a) What are the problems in direct mapping and associative mapping? How these problems are solved in set associative mapping? Describe. | 12 |
| 4 | (a) What is Cache Memory? Why do we need to use cache memory in our Computer System | 4 |
| | (b) Describe the operation techniques of Cache Memory | 4 |
| | (c) Explain the direct Mapping function of Cache Memory | 4 |
| 5 | (a) How CPU executes a Program? Give detailed description. | 8 |
| | (b) Differentiate among ROM, PROM, EPROM and EEPROM. | 4 |
| 6 | (a) Suppose, we read a word from main memory (where data bits are saved along with parity bits generated using the hamming code) and we get 101111100001. Is there any error here? If yes, show the corrected bits. | 12 |
| 7 | (a) What are the three types of hazards that can occur during instruction pipelining? Explain them briefly | 6 |
| | (b) How to deal with control hazards? Explain briefly | 6 |
| 8 | (a) Describe the process of write and read in a magnetic disc. | 8 |
| | (b) Why RAID is needed? Differentiate between RAID 1 and RAID 2. | 4 |



Jashore University of Science and Technology
Department of Computer Science & Engineering

B.Sc. Engineering Final Examination-2022
2nd Year 1st Semester, Session: 2019-2020

Course Code: CSE 2107

Time: 3.00 hours

Course Title: Numerical Analysis

Marks: 72

[N.B. Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks.]

1. a) Describe the Bisection method with proper figures. 04
b) Find the positive root of the equation $2x = 3 + \cos x$, by the bisection method, correct to the three decimal places. 04
c) Find a root of the equation $x^3 + x^2 - 1 = 0$ by using false position method, correct up to four decimal places. 04
2. a) Using Euler method, find an approximate value of y where $0 \leq x \leq 0.5$ and $h = 0.1$, given that $\frac{dy}{dx} = x^2 + y^2$ and $y(0) = 1$ 04
b) Find a real root of the following equations, by the Newton-Raphson method: 04
 $x^3 + 3x + 10 = 0$
c) Find a real root of the following equations, by the Secant method: $x^3 + 3x + 10 = 0$ 04
3. a) Given the following table. Calculate the $f(35)$ using forward difference 06

x	21	25	29	33	37
f(x)	18.47	17.81	17.10	16.34	15.51

b) Write a program for calculating the forward difference table of the above table. 06
4. a) Solve the following system of equations using the Gauss Elimination method: 06
 $2x_1 + 2x_2 + x_3 = 6$
 $4x_1 + 2x_2 + 3x_3 = 4$
 $x_1 + x_2 + x_3 = 0$
b) Write a program for above question. 06
5. a) What are the main objectives of numerical analysis? 02
b) Calculate the numerical integration using Simpson's rule of $f(x) = 3x^2 + 5x + 2$. Here, n , lower and upper limits are 8, -2 and 5, respectively. 06
c) Find the absolute error in $w = xy + z$ if $x = 3.35$, $y = 5.74$ and $z = 2.45$ 04

6. a) What are the main limitations of the bisection method?

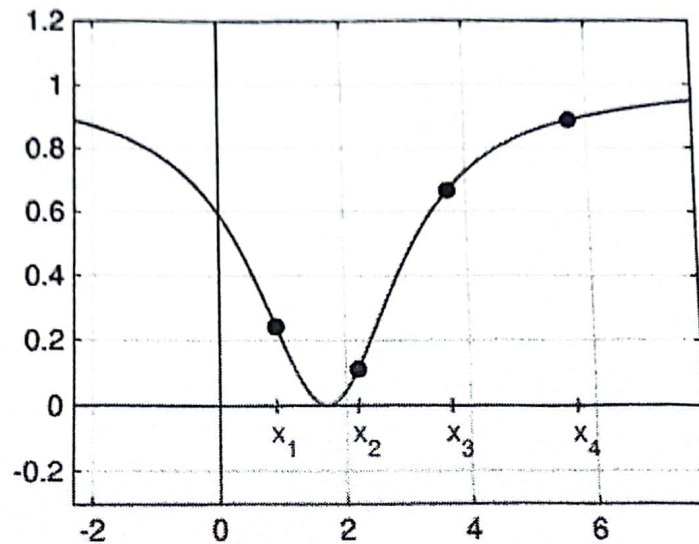
02

b) Compute the Taylor polynomial approximation to $\sin(x)$ expanded about the point $x = \pi/8$.

06

c) For the function $f(x)$ plotted below, use each of the four points x_1 , x_2 , x_3 and x_4 as the starting guess for Newton's method. For which point(s) do you expect the iteration to converge to the root shown? Also provide reasons for your answer.

04



7. a) Find the root of $f(x) = 3x + \sin(x) - ex$ using the Muller method.

06

b) Write a program for finding the root of the above equation.

06

8. a) Find the root of $f(x) = 3x + \sin(x) - ex$ using Hallie's method.

06

b) Write a program for finding the root of the above equation.

06

$$10/3 = 3\frac{1}{3}$$

$$10 - (3 \times 3)$$

$$= 10 - 9$$

$$= 1$$

$$15/2 = 7\frac{1}{2}$$

$$15 -$$

Department of Computer Science & Engineering

Jashore University of Science and Technology
2nd year 1st Semester, B.Sc. Engineering Final Examination, 2022

Course Title: Business Studies for Engineers
Time: 3 Hours

Course code: BUS 2101
Full marks: 72.0

(Answer any six set of questions from the followings)

1. a) Define administration. How administration differs from management? Describe. 1+4
b) What do you mean by organization? Briefly explain formal and informal organizational structure. 1+6
2. a) Differentiate between authority and responsibility. 4
b) What was Taylorism, and how did it give rise to Scientific Management Theory? 4
c) Describe about the major contributors to scientific management theory. 4
3. a) Suppose you have a private company to sell hardware products associated with vehicle. Your company is not making enough profit and you want to redesign a new structure for your organization. How you will do that? 6
b) What is span of control? Write the advantages and disadvantages of the narrow span of control and wide span of control. 1+5
4. a) Explain the meaning of economic laws according to Marshall and Robin. 3
b) Consider the nature of economic laws as behaviorist and Hypothetical and describe it. 4
c) Define engineering economics. Write down the characteristics of engineering economics. 1+4
5. a) Define management accounting. Differentiate between cost accounting and management accounting. 1+3
b) Explain the concept of management accounting. 4
c) Briefly describe the importance of management accounting in business sector. 4
6. a) The accounts of the Steelways Engineering Co. Ltd. show for 2010:

Materials used	\$1,80,000
Manual and machine labour wages directly chargeable	\$1,60,000
Works overhead expenditure	\$40,000
Embellishment and general expenses	\$19,000

- (i) Show the works cost and total cost, the percentage that the works overhead cost bears to the manual and machine labour wages and the percentage that the establishment and general expenses bear to the works cost. 4
- (ii) What price should the company quote to manufacture a machine which, it is estimated will require an expenditure of \$8,000 on materials and \$6,000 on wages so that it will yield a profit of 25% on the total cost or 20% on selling price. 3
- b) The following data relate to the manufacture of a standard product during the four-week period to June 30th, 2011: 5

Particulars	\$
Raw materials consumed	4,000
Wages	6,000
Machine hours worked	1,000
Machine hour rate	50 cents
Office overhead	20% on works cost
Selling overhead	6 cents per unit
Units produced	20,000
Units sold	18,000 @ Re. per unit

7. a) How important is motivation in the sense of organizational behavior? How motivation affects the performance of an employee in an organization? Explain. 6
- b) Suppose you have a Fast Track Courier service company and you are analyzing the cost factors and risk factors for your company. Which factors should you consider for this analysis? Describe 6
8. a) If you had to decide whether to continue making a component part or to begin buying the part from an overseas supplier, what quantitative and qualitative factors would influence your decision? 3
- b) Why do companies prepare budgets? 3
- c) How does managerial accounting differ from financial accounting? 2
- d) The administrator of Azalea Hills Hospital would like a cost formula linking the administrative costs involved in admitting patients to the number of patients admitted during a month. The Admitting Department's costs and the number of patients admitted during the immediately preceding eight months are given in the following table: 4

Month	Number of Patients Admitted	Admitting Department Costs
May	1,800	\$14,700
June	1,900	\$15,200
July	1,700	\$13,700
August	1,600	\$14,000
September	1,500	\$14,300
October	1,300	\$13,100
November	1,100	\$12,800
December	1,500	\$14,600

Required:

- i) Use the high-low method to estimate the fixed and variable components of admitting costs.

Express the fixed and variable components of admitting costs as a cost formula in the form $Y = a + bX$.

